

A freshly installed Gentoo Linux environment should be booting off your hard disk. Hooray!

How do I use Portage?

Portage is a package-management system as well as a package-distribution system; it shares some of the perspectives of the BSD ports system. Every package has its own plain-text file called *ebuilds* with package descriptions, source locations, dependencies and instructions on how to compile and install the package. Portage can be controlled using a Python script called *emerge*. In this section, I will introduce some basic usage of *emerge*.

Syncing the tree

Before searching or installing any of the packages, it is highly recommended to sync the tree first, as follows:

```
# emerge --sync
```



Note: Common Gentoo netiquette says you should not sync the tree more than once a day. Public Gentoo rsync mirrors are either run by the Gentoo infrastructure team, or by volunteers of the Gentoo community. The Gentoo rsync mirror consumes, on average, around 2.6 TB/month of bandwidth. Some rsync mirror administrators might ban your IP address if you sync more than twice in a single day. If you have multiple Gentoo systems on your LAN, try setting up an NFS share and exporting */usr/portage* to your clients, to save both your bandwidth, and the community's.

Searching for a package

```
$ emerge --search <pkgname>
```

You might notice that searching for a package with *emerge --search* takes quite some time. The *app-portage/eix* package indexes

packages, and allows faster queries. Install *eix*, build the database, and search the database of packages, as follows:

```
# emerge app-portage/eix
# eix-update
$ eix <pkgname>
```

Installing a package

```
# emerge <pkgname>
Calculating dependencies... done!
>>> Verifying ebuild manifests

>>> Emerging (1 of 1) app-category/<pkgname>-version
<..omitted..>
```

By default, *emerge* immediately starts downloading and compiling the software. To view the list of software, including dependencies, before actually installing them, add *--ask*. It will also ask you for confirmation. I usually use *-av*, which is short for *--ask* and *--verbose*. *--verbose* shows the *USE* flags, and summarises the size of downloads. For instance, *-av* will produce the following:

```
# emerge <pkgname> -av
```

These are the packages that would be merged, in order:

```
Calculating dependencies... done!
[build N ] app-category/<pkgname>-version
USE="edit gpm nls -X -samba -slang" 2,729 KB

Total: 1 package (1 new), Size of downloads: 2,729 KB

Would you like to merge these packages? [Yes/No] yes

>>> Verifying ebuild manifests

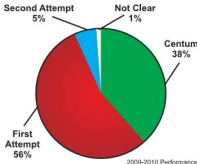
>>> Emerging (1 of 1) app-category/<pkgname>-version
<..omitted..>
```

Removing a package

```
# emerge -C <pkgname>
```

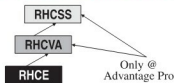
Again, run the same command with *--ask* and *--verbose* to show the


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 Vectra Technosoft Pvt. Ltd.
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Regd. Off: Wing 1 & 2, IV Floor,
 Jhaver Plaza, 1A, N.H. Road,
 Nungambakkam, Chennai - 34.
 Ph : 28263529 / 3530 / 3540
 Telefax : 28263527
 E-mail : enquiry@vectratech.in
 www.vectratech.in